

California Housing Price Prediction Using Linear Regression Report

1. Introduction

The main goal of this project is to predict house prices using the California Housing dataset. Machine Learning techniques can help estimate housing prices based on different factors such as income, house age, location, and population. In this project, Linear Regression was used as a baseline model to understand the relationship between these features and house prices.

2. Dataset Overview

The California Housing dataset contains information about houses from different regions of California. The dataset includes several features that may influence house prices, such as:

- Median Income (MedInc)
- House Age (HouseAge)
- Average Rooms (AveRooms)
- Average Bedrooms (AveBedrms)
- Population
- Average Occupancy (AveOccup)
- Latitude
- Longitude

The target variable for this project is **HousePrice**, which represents the value that the model aims to predict.

3. Exploratory Data Analysis (EDA)

Before building the model, the dataset was explored to better understand its structure and quality. Several checks were performed, including:

- Viewing dataset information using `info()`
- Examining statistical summaries using `describe()`
- Checking for missing values
- Analyzing feature relationships through a correlation heatmap

Key Observations

- The dataset contained no major missing values.
- Median Income showed the strongest positive correlation with house prices.
- Some features had weaker relationships with the target variable.
- The data appeared suitable for building a Linear Regression model.

4. Model Development

To build the prediction model, the dataset was divided into input features (X) and the target variable (y).

The following steps were performed:

1. Selected the relevant features and target variable.
2. Split the dataset into training and testing sets.
3. Applied StandardScaler to normalize the feature values.
4. Trained a Linear Regression model using the training data.
5. Generated predictions on the test dataset.

5. Model Evaluation

The model's performance was evaluated using three common regression metrics:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R^2 Score

Results

- MAE = [Insert Your Value]
- RMSE = [Insert Your Value]
- R^2 Score = [Insert Your Value]

Interpretation

The R^2 score indicates how well the model explains the variation in house prices. An R^2 value of approximately 0.61 means that the model explains about 61% of the variation in the target variable. While the model performs reasonably well, there is still room for improvement.

6. Visualizations

Several visualizations were created to better understand the data and evaluate model performance:

- Correlation Heatmap
- Actual vs Predicted Scatter Plot
- Residual Plot

These plots helped identify patterns in the data and assess the quality of the predictions made by the model.

7. Possible Improvements

Although Linear Regression provided satisfactory results, the model can be improved in several ways:

- Feature Engineering
- Outlier Detection and Removal
- Polynomial Regression
- Random Forest Regressor

- Gradient Boosting Regressor
- XGBoost Regressor

These advanced techniques may capture more complex relationships within the data and improve prediction accuracy.

8. Conclusion

This project demonstrated the use of Linear Regression for predicting California house prices. The dataset was explored, analyzed, and used to train a predictive model. The results showed that the model was able to capture a significant portion of the relationship between housing features and house prices. Overall, this project provided valuable hands-on experience in data preprocessing, exploratory data analysis, model training, and performance evaluation, while also highlighting opportunities for further improvement using more advanced machine learning algorithms.